# AI-Powered Multimodal Clinical Decision Support for Tropical and Infectious Disease Diagnosis Using Large Multimodal Models

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***Abstract*—Tropical infectious diseases pose significant diagnostic challenges, particularly in resource-limited settings where overlapping symptoms and fragmented clinical data increase uncertainty and delay treatment. Although recent advances in large language Models (LLMs) and large multimodal models (LMMs) offer potential for integrating multiple type of medical data (patient profile, labs, scans, etc.), current systems remain limited in domain-specific adaptation, multimodal reasoning, and interpretability, partly due to the lack of standardized datasets for tropical disease diagnosis. To address these gaps, we develop and comparatively evaluate three approaches for multimodal clinical decision support: (1) Multimodal Retrieval-Augmented Generation, (2) Fine-tuning LMMs with multimodal data for reasoning, and (3) a hybrid architecture combining both strategies. The models performances are assessed using automated evaluation framework based on RAGAS, and interpretability is enhanced through a LangGraph interface with step-level reasoning monitoring. Empirically, although overall accuracy remains limited, our results show that integrating both multimodal Retrieval-Augmented Generation (RAG) and fine-tuning substantially improves answer accuracy and reasoning capability, respectively. These findings highlight the potential of large multimodal models not as standalone diagnostic systems, but as assistive tools that can provide insightful reasoning and support differential diagnosis. Our code is publicly available on GitHub.**[1]

*Index Terms*—**Tropical Infectious Diseases, Multimodal Large Language Models, Medical Diagnosis, Retrieval-Augmented Generation, Fine-tuning,**

## I. Introduction

Tropical infectious diseases remain a major public health burden in numerous regions worldwide, particularly in regions with limited healthcare resources and specialist expertise. Early and accurate diagnosis is critical to reduce transmission, and improve patient outcomes.

Recent advances in Artificial Intelligence, especially Large Language Models (LLMs) and Large Multimodal Models (MLMMs), have demonstrated remarkable capabilities in understanding and reasoning over heterogeneous data types, including text, images, and structured medical data. When adapted to healthcare, these models offer the potential to integrate diverse clinical inputs and provide context-aware diagnostic support.

[1]https://github.com/fuisl/meditropical

However, for tropical and infectious diseases, current systems struggle to combine different data types and understand clinical context, making it difficult to provide accurate and trustworthy diagnostic reasoning. Additionally, the lack of high-quality multimodal datasets limits the training and testing of these models in real clinical settings.

Interpretability and trustworthiness are critical for clinical adoption, requiring AI-generated recommendations to be transparent and verifiable by clinicians to ensure patient safety and ethical standards.

For these reasons, there is a pressing need to develop robust multimodal AI systems integrating various data sources and types to enhance the diagnostic accuracy and reasoning capabilities for tropical infectious diseases. Multiple approaches, including Multimodal Retrieval-Augmented Generation (RAG), fine-tuning of pre-trained models on multimodal data, and hybrid strategies, should be implemented and rigorously evaluated to identify optimal solutions for clinical decision support in tropical infectious diseases.

The key contributions of this project include:

- Implementation and comparative evaluation of three approaches: Multimodal Retrieval-Augmented Generation (LightRAG [1] + RAGAnything [2]), domain-specific instruction fine-tuning with multimodal data (NVIDIA NeMo [3]), and a hybrid architecture combining both strategies.
- Construction of a multimodal benchmark dataset integrating clinical text, medical images, and structured patient data for tropical infectious disease diagnosis.
- Development of a fine-tuning dataset with structured diagnostic reasoning in USMLE [4] format.
- Evaluation using RAGAS [5] built-in and custom metrics.
- Interactive interface with step-level reasoning monitoring via LangGraph [6].

The remainder of this paper is organized as follows: Section II presents the problem statement, objectives, and research hypotheses. Section III details our methodology including data collection, data preparation, model building, evaluation, and interpretation. Section IV discusses our findings, interprets results, and outlines future research directions.

## II. Problem Statement and Objectives

The diagnosis process in the medical context is challenging due to its complexity and variability, together with the need for fast but accurate decision. However, as the general population rises, the medical resources and personnel are struggle with the increasing demand. This calls for a solution reliable enough for medical professionals to delegate some of their basic, repetitive tasks. Given the rises of Large Language Model (LLM) in recent years, this could be a solution to this problem.

However, existing LLM models for medical diagnosis still have significant limitations. Many models are trained on mono-modal data, mainly text, which is unrealistic and incomplete, given the nature of diagnostic decision-making process. Relying solely on text data limits the model's ability to capture the full clinical context, which risk making inaccurate or incomplete diagnoses. Furthermore, existing models are benchmarked on knowledge-heavy data [7], which contain a major portion of factual information. This benchmarking methods unreliably evaluate the model's reasoning ability.

Therefore, in this work, we seek to improve the model by integrating multi-modal capability, mainly images. We do this via multi-modal RAG, fine-tuning, and a hybrid approach combining both. We also seek to truly evaluate the model's reasoning ability by constructing a multi-modal reasoning benchmark, following MedCaseReasoning [8].

## III. Methodology

### A. Data Collection

*1) Data Sources:* We collect our data from various authoritative and reputable sources to ensure the factual correctness of our model, among which are textbooks, journals, and fact-sheets from worldwide institution.

For textbooks, we collect top-tier, standardized ones in the field, including *Manson's Tropical Infectious Diseases* (23rd edition) [9], *Manson's Tropical Diseases* (24th edition) [10], *Tropical and Parasitic Infections in the Intensive Care Unit* [11] (as part of the Perspectives on Critical Care Infectious Diseases Series), *Current Topics in Tropical Medicine* [12], and an e-book *The Diagnostic and Treatment of Tropical Diseases* [13]. We also crawl the fact-sheets from World Health Organization (WHO)'s official website. For the QA-set, we utilize the medbullet_op5 set from LangAGI-Lab on HuggingFace, which contains sampled questions from United States Medical Licensing Examination (USMLE), a well-known medical licensing examination in the United States. We also synthesize another set based on 93 clinical case reports provided by the lecturer *(from Clinical Cases in Tropical Medicine [14])* and 80 open-access case reports from the American Journal of Tropical Medicine and Hygiene (AJTMH). The data sources in our collection are summarized in Table I.

*2) Data Partition:* We partition the collected data according to their conventional use. For RAG, we use the textbooks and the fact-sheets to build the knowledge base. Meanwhile, we utilize the medbullet_op5 to perform instruction fine-tuning on our model. Finally, for benchmarking, we synthesize another set from the 173 case reports collected from multiple sources. Our partition is visualized in Figure 1.

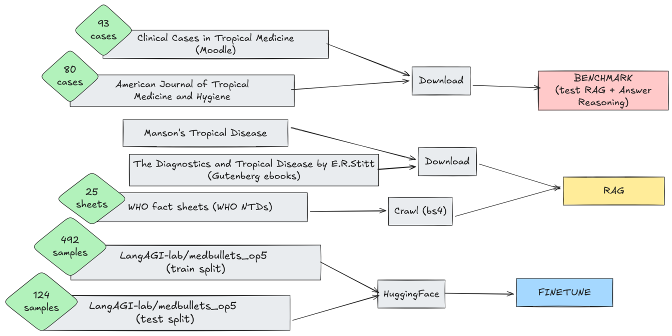



Figure 1. Data partition for different components

*3) Data Extraction:* We extract the data from its origin and form to further process it, as discussed in the next sections. Specifically, we directly download the textbooks from the corresponding official websites or digital libraries, as well as the clinical case reports from the journal. For the fact-sheets from WHO, we scrape the HTML content using web scraping tools. We obtain the QA-set from HuggingFace. Our basic extraction operations are shown in Figure 1

### B. Data Preparation

*1) Data Processing:* We preprocess the collected raw data to convert it into structured formats for our downstream use. In our case, only the fine-tuning and benchmarking datasets require extensive and systematic processing. Our processing steps for each dataset are shown in Figure 2.

*a) Fine-tuning Dataset Processing:* The dataset we collect from HuggingFace is a mono-modal multiple-choice Chain-of-Thought (CoT) dataset (as shown in Figure 12 and Figure 13), which contains a link to the corresponding question on a community-driven educational platform to prepare for the USMLE, as well as the question (as for *case description* or *case prompt*), the multiple choices and the correct choices, and the detailed reasoning for the correct answer. However, as we want our model to handle open-ended questions with a multi-modal input, we process the data to convert the question and reasoning to open-ended formats with medical image inputs. Specifically, we prompt-engineer based on MedCaseReasoning [8] paper which proven to be effective in generating high-quality diagnostic reasoning to rewrite the question and reasoning into an open-ended format. We then convert the pair into multi-modal input by crawling the relevant images via the

Table I
Overview of Data Sources

| Name | Volume | Type | Modality |
|---|---|---|---|
| Manson's Tropical Infectious Diseases (23rd Ed.) | 80 ch. / 1655 pp. | Textbook | Text + Images |
| Manson's Tropical Diseases (24th Ed.) | 84 ch. / 1634 pp. | Textbook | Text + Images |
| Tropical and Parasitic Infections in the ICU | 12 ch. / 247 pp. | Textbook | Text + Images |
| Current Topics in Tropical Medicine | 29 ch. / 578 pp. | Textbook | Text + Images |
| The Diagnostic and Treatment of Tropical Diseases (E-book) | 63 ch. / 754 pp. | Textbook | Text + Images |
| WHO Factsheets | 25 fact-sheets | Fact-sheet | Text |
| Medbullet Medical Exam Questions (medbullet_op5) | 160 QA pairs | QA-set | Text + Images |
| Clinical Cases in Tropical Medicine (Moodle) | 93 case reports | Case Report | Text + Images |
| AJTMH | 80 case reports | Case Report | Text + Images |

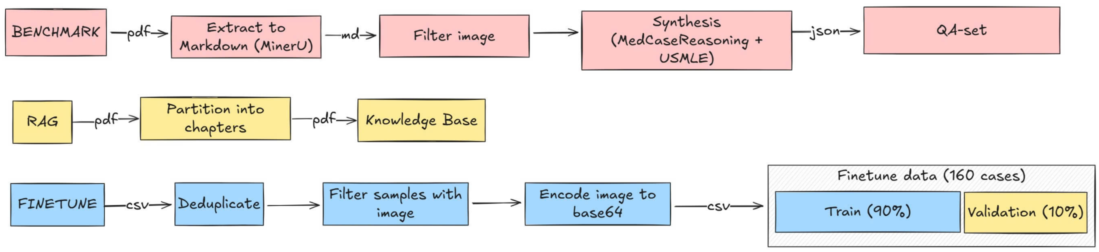



Figure 2. Data Processing for Different Sets

links provided using Firecrawl [15]. For convention, we append the based64 encoded image into an extra column in the dataset.

*b) Benchmarking Dataset Processing:* The case reports we collected are mostly in PDF, so firstly we extract it to markdown with MinerU [16]. We then filter the image paths, appending them immediately after the mention of the figure in the text. This is because MinerU utilizes Vision Language Model (VLM) to extract data based on layout, so the image paths are mixed in other parts. Similarly to the fine-tuning dataset, we also need to convert the case report into a question and reasoning pair following MedCaseReasoning [8] to handle multi-modal data since the original prompt deliberately omits other data modalities than text. Besides that, we prompt-tune for the model to regenerate the question and reasoning if needed to control the quality of the output. Finally, we devise the synthesizing pipeline and build a simple agentic loop with LangGraph [6] to iteratively generate the question and reasoning from the case report text using `gemma-3-27b-it` and `gemini-3`.

The loop is a simple state-machine process, with actions bound with the states, as shown in Figure 3. The loop contains two decision nodes, one to decide whether to process the case after evaluation, and one to decide whether to retry the case at the end.

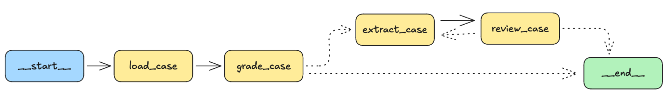



Figure 3. Data Extraction Graph

*2) Data Split:* We use our collection for RAG and benchmark in full. We only split the fine-tuning dataset into training and validation sets with 9:1 ratio.

### C. Model Building and Training

We implemented and evaluated three distinct approaches: multimodal RAG, multimodal instruction fine-tuning, and a hybrid combination. This section details the architecture, training pipeline, and technical implementation of each method

*1) Multimodal Retrieval-Augmented-Generation:*

*a) Architecture Overview:* RAG is essential in medical applications because it enables dynamic updates to medical knowledge bases without requiring model retraining, which is particularly important given the continuously evolving nature of medical information. RAG systems, including LightRAG [1], can ingest diverse medical data sources such as textbooks, case reports, and fact sheets, and store them in a database. When a query is given, the system retrieves relevant information from the knowledge base.

Unlike conventional RAG pipelines that primarily rely on vector similarity search, LightRAG has a graph-based approach. In our multimodal RAG setup, text and image inputs are handled jointly: text is used directly, while images are converted into textual descriptions with an adapter from RAGAnything [2], enabling both modalities to be indexed and retrieved in a unified pipeline. In LightRAG, entities represent meaningful medical concepts (e.g., diseases, symptoms, or drugs), while relationships capture the semantic connections between these entities (e.g., causes, treats, or associated with). By extracting entities and their relationships, LightRAG is able to retrieve more relevant information and uncover deeper contextual connections within the data.

*b) RAG Implementation Pipeline:* In a general RAG system, there are three distinct phases that define its operation: the data ingestion phase, the retrieval phase, and the generation phase. However, in our system, along side with LightRAG, we already have an Multimodal Large Language Model (MLLM) model available and want to retain control over context engineering. Therefore, in this work, we omit the final generation phase of LightRAG and focus solely on the first two phases.

i. Data ingestion phase: As illustrated in Fig. 4, uploaded data sources are separated into text and image inputs, with images converted to text via the RAGAnything adapter. The resulting text is chunked, entities and relationships are extracted to build the knowledge graph, and the artifacts are stored in a vector database and a key–value store.

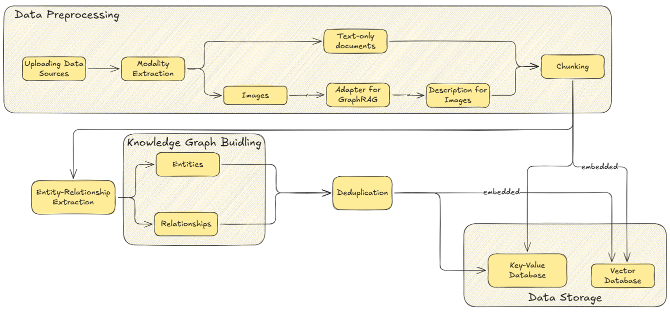



Figure 4. Overview of data ingestion process

ii. Retrieval phase: As illustrated in Fig. 5, when a user query is given, the retrieval process operates simultaneously in three modes. In the Naive mode, vector similarity is computed directly between the embedded user query and embedded data chunks to retrieve the most similar chunks. In the Local and Global modes, embeddings of entities and relationships are compared with the embedded user query, after which the most similar entities and relationships are mapped to their corresponding data chunks. Finally, data chunks retrieved from all three modes are aggregated and reranked based on their relevance to the query before being returned.

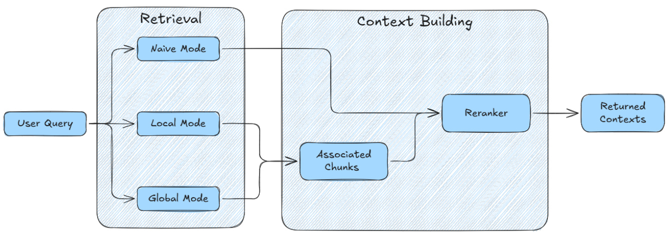



Figure 5. Overview of retrieval process

*2) Fine-tuning Pipeline:*

*a) Overview Fine-tuning:* Fine-tuning adapts a pre-trained large language model to a specific task or domain using supervised data. In this work, we employ instruction fine-tuning with multimodal data, training the model on our curated multimodal question–answer dataset. The objective is to teach the model to perform clinical diagnosis following the format proposed in the MedCaseReasoning paper, which includes both the final diagnosis and the corresponding reasoning. Through this process, we aim to improve diagnostic accuracy while producing more coherent and plausible reasoning, which improves the interpretability in sensitive medical contexts. Fine-tuning is implemented using NVIDIA NeMo [3], a framework optimized for efficient GPU-based training.

*b) Fine-tuning Pipeline:* As illustrated in Fig. 6, the fine-tuning pipeline in NeMo generally has five steps.

i. Step 1: We take in the benchmark dataset as a csv file.

ii. Step 2: We convert the dataset into NeMo-compatible formats, particularly WebDataset (WDS) shards [17] and ingest it through NeMo's data loader.

iii. Step 3: We set training configuration and select model components (e.g., attention layers) for LoRA [18] fine-tuning according to the provider's recipe.

iv. Step 4: The fine-tuning process is executed using NeMo's trainer module and monitored using Weight and Biases (Wandb) [19].

v. After training, the learned LoRA adapters or fine-tuned checkpoints are exported. During inference later, we can serve the resulting model using vLLM [20].


Figure 6. Overview of fine-tuning pipeline

*c) Fine-tuning Configuration:* We fine-tune the `Gemma 4B Instruct` model using a parameter-efficient approach based on Low-Rank Adaptation (LoRA) [18], optimized with the Adam optimizer [21]. The LoRA configuration uses a rank of $r = 4$ and a scaling factor of $\alpha = 16$. Training is performed for a maximum of 1,000 steps with early stopping applied using a patience of 10 steps. The learning rate is set to $1 \times 10^{-5}$, with a weight decay of 0.01.

The dataset consists of 109 cases and is split into training and validation sets using an 9:1 ratio. All experiments are conducted on a single NVIDIA RTX 4090 GPU.

*3) Hybrid Combination Approach and User Interface:*

*a) Combination Overview:* To leverage the strengths of both methods, we integrate them into a unified pipeline using LangGraph. As illustrated in Fig. 7, which is visualized through the built-in LangGraph CLI interface, the overall workflow consists of three main steps: case extraction, model calling, and result export. Within the model invocation step, a conditional mechanism allows users to select among four operating modes: a baseline model only, a model augmented with RAG, a fine-tuned model, and a model enhanced with both RAG and fine-tuning.

During inference, the user provides a clinical question along with associated images, either as local file paths or in Base64-encoded format. The RAG system first retrieves relevant contextual information based on the question and then concatenates the question, the images (with local files loaded and encoded into Base64), and the retrieved context into a single model input. The pipeline produces a concise output that directly answers the question, such as a diagnosis or symptom, followed by the corresponding reasoning. The system prompt and the concatenated prompt format used during inference are provided in the Appendix VI.

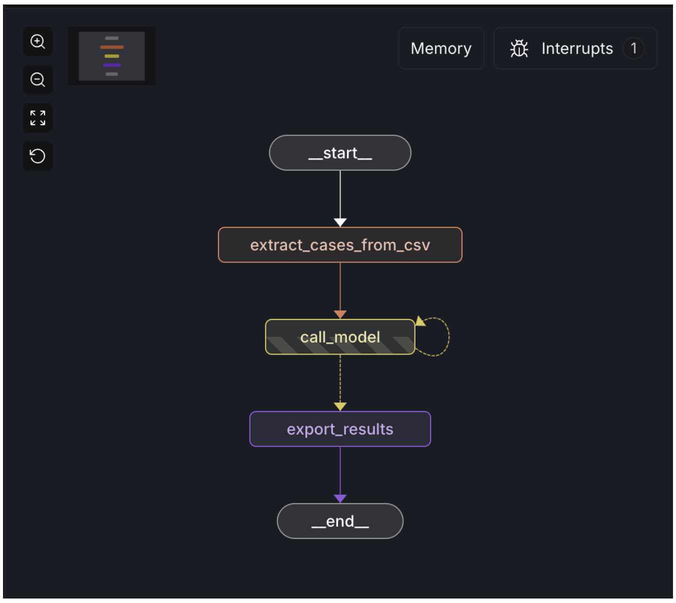



Figure 7. Overview of combined pipeline

*b) User Interface:* To simulate real usage and bring the application closer to its intended end users, such as patients and clinicians, we developed an interface based on the LangGraph CLI that enables real-time chat with support for image attachments. Although the diagnostic process is currently limited to single-turn interactions, the interface abstracts away technical details and allows users to focus on the system's functionality. A demonstration conversation between a human user and the AI system is shown in Fig. 8. The AI first presents a concise answer, followed by the corresponding reasoning.

## D. Evaluation Framework

*a) Evaluation Overview:* With the improving generalization capabilities of LLMs, a growing trend in evaluation is the use of the *LLM-as-judge* paradigm [22]. Instead of relying on domain-specific human experts for manual evaluation, an LLM judge automatically grades model outputs based on predefined prompts and evaluation metrics. While this approach may be less reliable in

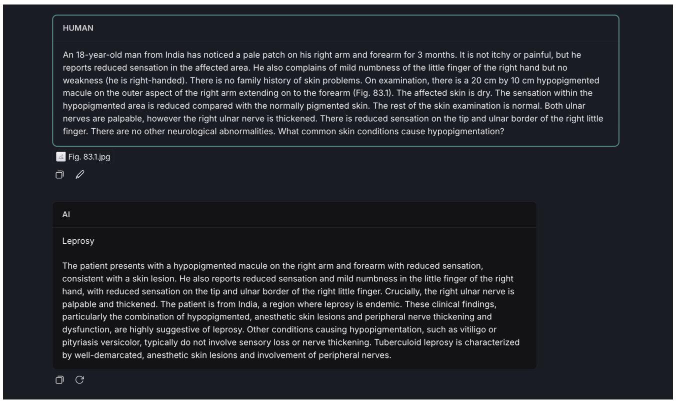



Figure 8. Overview of user interface

terms of absolute correctness and trustworthiness compared to expert human evaluation, well-designed metrics and instructions can provide high flexibility, fast evaluation speed, and rapid insights into system performance. In this work, we adopt the RAGAS framework to conduct LLM-as-judge evaluations using both standardized built-in metrics and custom-defined metrics.

*b) Evaluation Metrics:* The input JSON used for evaluation contains a subset of the following fields, depending on the metrics being applied:

- `question`: the test case
- `ground_truth`: the ground-truth final diagnosis
- `ground_truth_reasoning`: the ground-truth reasoning
- `contexts`: contexts retrieved by the RAG system
- `answer`: the model's final diagnosis
- `answer_reasoning`: the model's reasoning

However, the RAGAS framework internally supports only four standardized fields:

- `user_input`
- `response`
- `retrieved_context`
- `reference`

As a result, we decompose the evaluation pipeline into three separate stages to assess different aspects of system performance: RAG, answer, and reasoning.

The summary of all the used metrics is illustrated in Table II.

*c) Evaluation Settings:* We conduct evaluations on retrieval, final answers, and reasoning using a benchmark dataset comprising 78 cases. The LLM-as-judge selected for this evaluation is `MedGemma-4b-Instruct`. Although common practice recommends using a judge model that is different from the evaluated models and ideally from a different model family to mitigate bias, the requirement for strong medical domain knowledge motivates our choice of a model pre-trained on medical datasets to ensure reliable evaluation.

For each test case, contextual information is retrieved once and then shared across all model settings during

Table II
Summary of Evaluation Metrics

| Metric Name | Purpose | Meaning | Provider |
|---|---|---|---|
| Context Recall | RAG Evaluation | Measures whether all claims in *ground_truth* and *ground_truth_reasoning* are supported in the retrieved *contexts* | RAGAS |
| Context Precision | RAG Evaluation | Measures whether the retriever ranks relevant chunks higher than irrelevant ones in the retrieved *contexts* | RAGAS |
| Faithfulness | Answer Evaluation | Measures whether the *answer* is factually grounded in the provided *contexts* | RAGAS |
| Answer Relevance | Answer Evaluation | Measures whether the *answer* is relevant to the given *question* | Custom |
| Answer Accuracy | Answer Evaluation | Measures whether the *answer* matches the *ground_truth* | Custom |
| Reasoning Recall | Reasoning Evaluation | Measures whether the *answer_reasoning* covers all key elements of the *ground_truth_reasoning* | Custom |

inference to ensure a fair comparison while reducing computational overhead. RAG system employs the `bge-m3` embedding model and `MedGemma-4b` as the retrieval model. Given a text-only query, it retrieves the top 20 most relevant data chunks. Each evaluated model is expected to produce a single final diagnosis along with the corresponding reasoning. In total, we evaluate six diagnostic settings using three models, including `MedGemma-4b`, `Gemma-3-4b-it`, and `Gemma-3-4b-it-finetuned`, each evaluated in two modes: with RAG and without RAG. The RAG-supported configurations are denoted by the suffix "+ RAG" in the model name.

*d) Evaluation Results:* We first evaluate the performance of the multimodal RAG system. As shown in Table III, LightRAG+RagAnything achieves a high Context Recall score of approximately 0.9, indicating effective retrieval of relevant information. However, the Context Precision score remains moderate, at below 0.6, suggesting that retrieved contexts are not optimally ranked by relevance.

Table III
Multimodal RAG (M-RAG) Retrieval Performance. For all metrics, higher scores indicate better performance with minimum score is 0 and maximum score is 1.

| Method | Output | Context Recall | Context Precision |
|---|---|---|---|
| M-RAG | Top-20 Chunks | 0.8932 | 0.5651 |

Next, we evaluate the answer and reasoning performance of each model setting. The detailed scores, reported in Table IV, show that all settings achieve relatively similar results on the Faithfulness and Answer Relevance metrics, with the maximum difference between the best- and worst-performing models being approximately 0.1 for each metric. In contrast, Answer Accuracy and Reasoning Recall exhibit a wider range of performance across different settings. These results are further visualized using bar charts for more exploratory analysis in Fig. 9 and Fig. 10.

### E. Results Interpretation and Analysis

*a) Strengths:* Upon reviewing the results, several promising findings were observed:

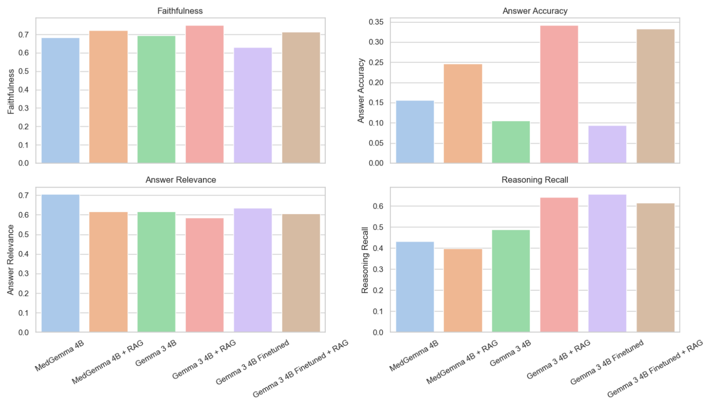



Figure 9. Comparison of each configuration across the four evaluation metrics

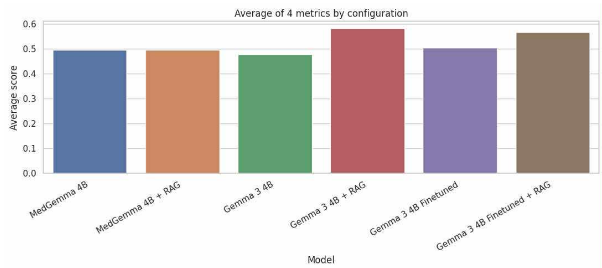



Figure 10. Average performance across the four metrics by configuration

- Among models evaluated without RAG contexts, `MedGemma-4b-Instruct` performs best, achieving an accuracy nearly 1.5× higher than the other two models. This result is expected, as MedGemma is extensively fine-tuned on medical datasets, giving it stronger internal medical knowledge compared to the other models.
- LightRAG retrieves high-quality data chunks, as evidenced by a Context Recall score of approximately 0.9 out of 1. This demonstrates not only the effectiveness of the RAG system but also that the constructed knowledge base is sufficient to answer questions from the benchmark dataset.

Table IV
PERFORMANCE COMPARISON OF MODELS WITH AND WITHOUT RAG. THE BEST AND SECOND-BEST RESULTS FOR EACH METRIC ARE HIGHLIGHTED IN **BOLD** AND UNDERLINED, RESPECTIVELY. FOR ALL METRICS, HIGHER SCORES INDICATE BETTER PERFORMANCE WITH MINIMUM SCORE IS 0 AND MAXIMUM SCORE IS 1.

| Models | Output | Faithfulness | Answer Accuracy | Answer Relevance | Reasoning Recall |
|---|---|---|---|---|---|
| `medGemma-4b` | Answer + Reasoning | 0.6837 | 0.1562 | **0.7051** | 0.4333 |
| `medGemma-4b` + RAG | Answer + Reasoning | <u>0.7225</u> | 0.2466 | 0.6167 | 0.3979 |
| `gemma3-4b-it` | Answer + Reasoning | 0.6944 | 0.1053 | 0.6167 | 0.4887 |
| `gemma3-4b-it` + RAG | Answer + Reasoning | **0.7512** | **0.3421** | 0.5859 | <u>0.6434</u> |
| `gemma3-4b-it-finetuned` | Answer + Reasoning | 0.6313 | 0.0938 | <u>0.6359</u> | **0.6564** |
| `gemma3-4b-it-finetuned` + RAG | Answer + Reasoning | 0.7130 | <u>0.3333</u> | 0.6051 | 0.6165 |

- Both RAG and fine-tuning achieve their intended purposes:
  - For RAG, the goal is to retrieve relevant domain-specific knowledge to support diagnosis. The results show that RAG successfully fulfills this role, as all three models exhibit substantial performance improvements. In particular, Gemma-3-4b-it and Gemma-3-4b-it-finetuned improve their accuracy by more than 3× compared to their non-RAG counterparts. `MedGemma-4b-Instruct` shows a more modest improvement of over 1.5×, which we attribute to its stronger reliance on internal knowledge, which primarily covers general medical cases rather than infectious tropical diseases.
  - For fine-tuning, the objective is to teach the model to generate plausible and trustworthy reasoning to justify its final diagnosis. The results indicate that fine-tuning significantly improves the reasoning capability of Gemma-3-4b-it. The fine-tuned models achieve among the highest reasoning recall scores, closely matching the ground-truth reasoning traces. Additionally, Gemma-3-4b-it with RAG also achieves high reasoning recall, likely due to high-quality retrieved knowledge chunks.

*b) Limitations:* Despite the positive results, several limitations are identified:

- The RAG system exhibits relatively low context precision (below 0.6), indicating that retrieved data chunks are not optimally ranked by relevance, despite their high quality as shown by Context Recall. This limitation arises because the reranker module was disabled during retrieval due to resource constraints. This metric is expected to improve with an appropriate reranking model in future experiments.
- The overall accuracy across all settings remains low (below 0.4), reflecting the high difficulty of the benchmark task. The models must reason over patient context, retrieved medical knowledge, and images. Additionally, the strict evaluation metric requires an almost exact match to the ground-truth diagnosis, with no credit for partial answers. Given the high reasoning recall scores, we believe that evaluating differential diagnosis performance (i.e., top-$k$ predictions with $k > 1$) would better reflect the models' true potential, as their reasoning is largely aligned with correct clinical pathways.

*c) Conclusion:* Overall, our experiments indicate that:

- Both RAG and fine-tuning successfully achieve their intended objectives, improving model accuracy and reasoning capability, respectively when working with multimodal medical data.
- Although the models do not perform reliably as independent diagnostic systems due to low exact-match accuracy, their strong reasoning abilities suggest they are well-suited as clinical decision-support tools for generating differential diagnoses.

## IV. CONCLUSION AND OUTLOOK

This work addresses key challenges in clinical decision support for tropical and infectious diseases, where existing systems remain limited by insufficient multimodal integration, poor generalization to tropical infectious diseases, and lack of interpretability.

Our contributions to overcome these limitations include implementation and systematic comparative evaluation of three multimodal clinical decision-support approaches: Multimodal RAG (LightRAG + RAGAnything), domain-specific fine-tuning using NVIDIA NeMo, and a hybrid architecture integrating both strategies. We further construct a standardized multimodal benchmark dataset and develop a unified fine-tuning dataset with structured diagnostic answers and step-by-step reasoning aligned with a USMLE-inspired MedCaseReasoning workflow. In addition, we design a rigorous evaluation framework leveraging RAGAS to assess diagnostic accuracy and reasoning quality, and implement an interactive LangGraph-based interface with step-level reasoning monitoring to enhance transparency and interpretability.

When interpreting the results, we can observe that while the implemented models are not yet reliable as fully autonomous diagnostic clinical decision-support systems, their strong reasoning alignment and significant performance improvements through RAG and fine-tuning integration highlight their meaningful clinical potential. Specifically, the combination of external knowledge retrieval and enhanced reasoning capability positions these models as promising clinical decision-support tools, especially for generating structured and clinically relevant differential diagnoses. Future works should focus on further improving context precision and overall diagnostic accuracy by integrating an effective reranking module into the retrieval pipeline. We additionally recommend expanding and refining the benchmark dataset to increase the disease diversity and clinical realism. Subsequently, exploring and adopting more clinically meaningful evaluation metrics, such as top-k differential diagnosis accuracy, will better reflect real-world diagnostic performance.

## V. Acronyms

| | |
|---|---|
| **RAG** | Retrieval-Augmented Generation |
| **CoT** | Chain-of-Thought |
| **USMLE** | United States Medical Licensing Examination |
| **AJTMH** | American Journal of Tropical Medicine and Hygiene |
| **WHO** | World Health Organization |
| **ICU** | Intensive Care Unit |
| **QA** | Question-Answering |
| **AI** | Artificial Intelligence |
| **MLLM** | Multimodal Large Language Model |
| **LLM** | Large Language Model |
| **VLM** | Vision Language Model |

## VI. Appendix

### A. Initial Data Curation Approach

In the initial phase of our project, our benchmark dataset curation process also followed 2 stages similar to those described in Section III-B. However, instead of using MinerU for data extraction to Markdown format and LangGraph for building an agentic loop to generate the QA sets, we followed a more manual and more inefficient approach. For data extraction, we manually converted the PDF case reports to markdown using PyMuPDF. However, this approach often led to formatting issues as well as tables and images being misplaced or lost, which required additional manual intervention to correct. This limitation motivated us to research other approaches for handling this task much more effectively, leading us to MinerU. This tool significantly improved the quality of our extracted data and reduced the time spent on manual corrections. Regarding the generation of the QA sets, we initially employed a more straightforward prompting method using *gemini-2.5-flash* without an agentic loop. This approach, however, often resulted in incomplete or less coherent question and reasoning pairs, necessitating multiple rounds of manual review and editing. After exploring alternative methods, we discovered LangGraph, the framework that allowed us to generate the QA sets with structured output. This method not only improved the coherence and completeness of the generated content but also streamlined the overall process, reducing the need for extensive manual intervention.

### B. Task Allocations

We divide our group into two teams, the AI team and Data team, and allocate the tasks to each group accordingly. The individual group will decide on further task allocations within itself. For the AI team, Duong is responsible for the model building and training, while Minh focuses on model evaluation. Meanwhile, the Data team discharges its tasks collaboratively, with both members contributing to different aspects of data collection and preprocessing, often overlapping each other. Our task allocations are summarized in Table V.

| Member | Task | Team |
|---|---|---|
| Duong | Build RAG knowledge base<br>Fine-tune model<br>Combine methods | AI |
| Minh | Evaluate model<br>Interpret result<br>Combine methods | AI |
| Phuc | Synthesize benchmark<br>Prepare fine-tuning data | Data |
| Xuan | Prepare data for knowledge base<br>Experiment with benchmark data synthesis | Data |

Table V
Task Allocations of Group Members

### C. Knowledge Graph

LightRAG provides a built-in user interface that offers an overview of the entire knowledge graph. When hovering over a node, detailed information is displayed, including the node's identifier, label, degree (i.e., the number of neighboring nodes), properties, and associated relationships. An overview of the knowledge graph along with a sample node is illustrated in Fig. 11.

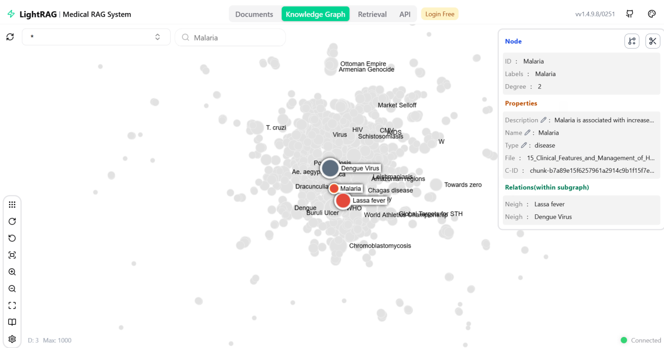



Figure 11. Overview of the knowledge graph and a sample node

### D. Prompts for Diagnostic Inference

To maintain consistency with the benchmark answer and reasoning style, we took inspiration from the MedCaseReasoning prompt during inference and adopt the special tags <think> and <answer>. This prompt design ensures that the model produces separate outputs for reasoning and final answers while generating both within a single inference call.

```
MEDCASE_SYSTEM_PROMPT = """You are a medical reasoning
     assistant.

Follow these rules STRICTLY:
1. First, reason internally and output it ONLY inside
     <think>...</think>.
2. Then output ONLY the final answer inside
     <answer>...</answer>.
3. Do NOT include the final answer inside the
     reasoning.
4. Base reasoning only on provided context, images,
     and established medical facts.
5. Explicitly state uncertainty if evidence is
     insufficient.
6. Use concise, professional clinical language.
7. Output plain text only. No markdown.

You MUST follow the output template exactly.
"""
```

Listing 1. System Prompt Used for Medical Reasoning

```
f"""
----------------------------------------
CASE PRESENTATION
----------------------------------------
Question:
{question}

Retrieved Context:
{context_block if context_block else "None"}

Images:
{image_block if image_block else "None"}

----------------------------------------
OUTPUT TEMPLATE
----------------------------------------
<think>
...your internal reasoning...
</think>
<answer>
...final answer only...
</answer>

----------------------------------------
EXAMPLES
----------------------------------------
{FEW_SHOT_EXAMPLES}
"""
```

Listing 2. Inference-Time Prompt Formatting Template

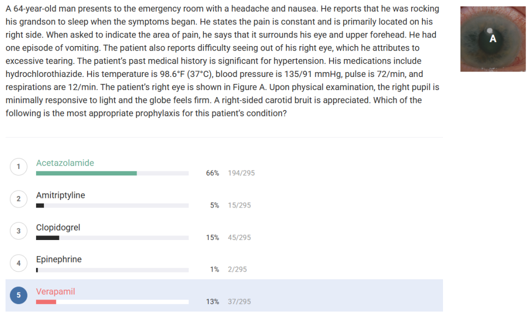
### *E. Data Samples*

We showcase samples from our fine-tuning dataset.

We here display a sample Question-Answering (QA) pair with reasoning from the fine-tuning dataset, via screenshot of the web. As can be seen here, the reasoning is eliminative, being strictly for multiple-choice format.

We also show a sample from our benchmark dataset in JSON format below.

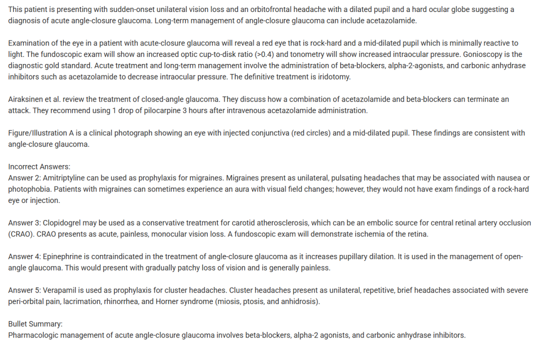

```
{
  "source":"case_33",
  "case_prompt":"A 53-year-old Malawian
      man presents to a local hospital
      with a productive cough and whitish
      sputum for 3 months. He also reports
```

A 64-year-old man presents to the emergency room with a headache and nausea. He reports that he was rocking his grandson to sleep when the symptoms began. He states the pain is constant and is primarily located on his right side. When asked to indicate the area of pain, he says that it surrounds his eye and upper forehead. He had one episode of vomiting. The patient also reports difficulty seeing out of his right eye, which he attributes to excessive tearing. The patient's past medical history is significant for hypertension. His medications include hydrochlorothiazide. His temperature is 98.6°F (37°C), blood pressure is 135/91 mmHg, pulse is 72/min, and respirations are 12/min. The patient's right eye is shown in Figure A. Upon physical examination, the right pupil is minimally responsive to light and the globe feels firm. A right-sided carotid bruit is appreciated. Which of the following is the most appropriate prophylaxis for this patient's condition?



Figure 12. Question-Answer pair with image from medbullet_op5 dataset



This patient is presenting with sudden-onset unilateral vision loss and an orbitofrontal headache with a dilated pupil and a hard ocular globe suggesting a diagnosis of acute angle-closure glaucoma. Long-term management of angle-closure glaucoma can include acetazolamide.

Examination of the eye in a patient with acute-closure glaucoma will reveal a red eye that is rock-hard and a mid-dilated pupil which is minimally reactive to light. The fundoscopic exam will show an increased optic cup-to-disk ratio (>0.4) and tonometry will show increased intraocular pressure. Gonioscopy is the diagnostic gold standard. Acute treatment and long-term management involve the administration of beta-blockers, alpha-2-agonists, and carbonic anhydrase inhibitors such as acetazolamide to decrease intraocular pressure. The definitive treatment is iridotomy.

Airaksinen et al. review the treatment of closed-angle glaucoma. They discuss how a combination of acetazolamide and beta-blockers can terminate an attack. They recommend using 1 drop of pilocarpine 3 hours after intravenous acetazolamide administration.

Figure/Illustration A is a clinical photograph showing an eye with injected conjunctiva (red circles) and a mid-dilated pupil. These findings are consistent with angle-closure glaucoma.

Incorrect Answers:
Answer 2: Amitriptyline can be used as prophylaxis for migraines. Migraines present as unilateral, pulsating headaches that may be associated with nausea or photophobia. Patients with migraines can sometimes experience an aura with visual field changes; however, they would not have exam findings of a rock-hard eye or injection.

Answer 3: Clopidogrel may be used as a conservative treatment for carotid atherosclerosis, which can be an embolic source for central retinal artery occlusion (CRAO). CRAO presents as acute, painless, monocular vision loss. A fundoscopic exam will demonstrate ischemia of the retina.

Answer 4: Epinephrine is contraindicated in the treatment of angle-closure glaucoma as it increases pupillary dilation. It is used in the management of open-angle glaucoma. This would present with gradually patchy loss of vision and is generally painless.

Answer 5: Verapamil is used as prophylaxis for cluster headaches. Cluster headaches present as unilateral, repetitive, brief headaches associated with severe peri-orbital pain, lacrimation, rhinorrhea, and Horner syndrome (miosis, ptosis, and anhidrosis).

Bullet Summary:
Pharmacologic management of acute angle-closure glaucoma involves beta-blockers, alpha-2 agonists, and carbonic anhydrase inhibitors.

Figure 13. Reasoning from medbullet_op5 dataset


```
      night sweats and some weight loss,
      which he is unable to quantify. Two
      months prior, he was treated with
      presumptive antimalarials and
      amoxicillin for 5 days without
      improvement. He denies TB contacts,
      is a non-smoker, and has no
      significant past medical history. He
      is a subsistence farmer, married
      with five well children.\n\nOn
      examination, his temperature is
      38.8\u00b0C (101.8\u00b0F), blood
      pressure is 100/80 mmHg, pulse is 88
      bpm, and oxygen saturation is 97% on
      ambient air. He appears mildly
      anaemic. Oral examination reveals no
      oral thrush, Kaposi\u2019s sarcoma,
      or oral hairy leukoplakia. His chest
      is clear with normal heart sounds.
      There is no lymphadenopathy, and his
      liver and spleen are not
      enlarged.\n\nLaboratory results
      show: WBC 3.2 x10\u00b0/L (reference
      4-10), Haemoglobin 9.8 g/dL
      (reference 13-15), MCV 90 fL
      (reference 80-98), Platelets 305
      x10\u00b0/L (reference 150-350), CD4
      count 54 cells/\u03bcL (reference
```

```
        500-1200), Sputum for AFB 2 x
        negative (reference Negative),
        Malaria RDT Negative (reference
        Negative), Thick smear for
        Plasmodium spp. Negative (reference
        Negative).\n\nHis chest radiograph
        on admission shows a prominent hilar
        region bilaterally but is otherwise
        normal (Fig. 33.1).\n\nWhat is your
        suspected diagnosis?",
    "final_diagnosis":"Tuberculosis",
    "reasoning":"The diagnostic process
        begins with recognizing the
        patient's travel history to
        Cameroon, an endemic area for
        several parasitic infections. The
        key finding of eosinophilia, coupled
        with transient subcutaneous
        swellings, immediately points
        towards a parasitic etiology.
        Considering the patient's location,
        filarial infections become a primary
        concern. The cardiac symptoms, while
        potentially unrelated, necessitate
        investigation for possible parasitic
        involvement of the heart. The
        differential diagnosis must include
        various filarial infections known to
        cause these symptoms, such as
        loiasis, lymphatic filariasis, and
        other less common parasitic
        diseases. Further investigations,
        such as microscopic examination of
        blood samples for microfilariae, are
        crucial to confirm the diagnosis and
        identify the specific parasite
        involved.",
  }
```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